2.3 Focus of a Parabola

Name _____ Period ____ Date _____

Algebra II 2.3 (30-minute plan) · Student sheet

I can…

- identify the focus, directrix, and axis of symmetry of a parabola.
- write equations of parabolas from the focus, directrix, or vertex.
- solve real-life problems involving the focus of a parabola.

Vocabulary · fill in the missing word

Parabola — the set of all points _____ distant from a point (the focus) and a line (the directrix)

Focus — the fixed point inside the _____

Directrix — the fixed line outside the _____

Vertical axis (vertex at origin) — $y = (\frac{1}{4p})x^2$: focus (0, p), _____ $y = -p$

Horizontal axis (vertex at origin) — $x = (\frac{1}{4p})y^2$: focus (p, 0), _____ $x = -p$

Part 1 · Notes and examples (fill in each note, then work the example)

Parabola: Every point on a parabola is the same distance from the _____ and the _____.

Vertical axis: $y = (1/(4p))x^2$: focus _____, directrix _____.

Example 1

Identify the focus, directrix, and axis of symmetry.

a. $y = -(\frac{1}{12})x^2$ b. $3x^2 - 24y = 0$

Horizontal axis: $x = (1/(4p))y^2$: focus _____, directrix _____.

Example 2

Identify the focus, directrix, and axis of symmetry of $x = \frac{1}{4}y^2$. Which way does it open?

Part 2 · Partner practice

Partner practice

Identify the focus and directrix of $y = (\frac{1}{16})x^2$.